

AK VisionFlow Owner Android App

Phase 4 — AI Business Advisor & Daily AI Recommendations

Phase 4 adds the intelligence layer that interprets the business data already available in VisionFlow. The goal is not to create a generic chatbot, but an AI business advisor that identifies meaningful changes, explains likely causes, highlights priorities, and recommends what the owner should review or consider.

1. Phase Objective

Create an AI-powered owner advisory system that continuously evaluates approved business KPIs, trends, alerts and operational signals, then produces concise, evidence-based insights for the owner. The Android app remains read-only; AI recommendations do not directly execute business actions.

2. AI Advisor Core Questions

- **What happened?** Identify important changes in sales, profit, expenses, receivables, inventory and other KPIs.
- **Why might it have happened?** Compare relevant dimensions such as branch, category, product, staff, discounts or payment patterns.
- **What needs attention?** Rank the most important issues rather than presenting dozens of observations.
- **What should the owner consider?** Give practical, data-supported recommendations and suggested areas to review.

3. Daily AI Business Advice

Once per day, the system generates a concise owner briefing based on the latest available business data. The message should focus on the highest-value insight of the day and may include supporting evidence.

Example:

Today's Business Advice

Sales increased 14% this week, but profit increased only 4%. The largest difference is concentrated in the Optical category, where average discounts increased. Review high-discount orders and margin performance today.

4. AI Insight Types

Insight Type	Purpose
Performance	Explain significant sales, profit or operational performance changes.
Anomaly	Identify unusual behavior compared with historical or expected patterns.
Risk	Highlight conditions that could negatively affect cash flow, margin, stock or customers.
Opportunity	Identify positive trends or areas where the business may benefit from further attention.
Trend	Explain sustained increases/decreases rather than isolated daily changes.
Recommendation	Suggest a specific area, report, category, branch or metric the owner should review.

5. AI “Needs Attention” Engine

The app should provide a prioritized list of the most important issues discovered by the AI. This becomes the owner's quick decision-support layer.

Example:

1. Receivables increased 21% — review overdue accounts.
2. Profit margin fell 4.2% — investigate discounting/cost changes.
3. 8 high-value products are out of stock — review replenishment.
4. One branch is 18% below its normal sales trend.

6. Evidence-Based AI

Every AI insight should be grounded in actual VisionFlow metrics. The system should provide the underlying figures, comparison period and relevant dimensions to the AI rather than asking the model to infer facts from incomplete raw data.

- AI must distinguish facts from interpretations.
- Use exact KPI values and comparison periods where available.
- Avoid unsupported claims about causes.
- When the cause is uncertain, explicitly say that it is a likely explanation or requires review.
- AI recommendations should be traceable to the business signals that triggered them.

7. AI Confidence & Guardrails

The advisor is a decision-support feature, not an autonomous business operator. It must not silently make changes to products, stock, pricing, payments, accounting or other records.

- Do not invent missing business data.
- Do not present correlation as proven causation.
- Do not recommend actions outside the available evidence.
- Use conservative language when data is incomplete.
- Clearly separate financial/business observations from recommendations.
- Never execute an operational action directly from an AI recommendation.

8. AI Advisor Screens

Screen	Function
AI Advisor Home	Latest AI insights, priority items and daily advice.
Daily Briefing	Today's summarized business advice with supporting metrics.
Insight Detail	Explanation, evidence, comparisons and suggested review area.
Needs Attention	Prioritized issues requiring owner attention.
AI History	Previous daily recommendations and insights for reference.

9. Daily AI Notification

Phase 3 provides the notification infrastructure. Phase 4 supplies the intelligence behind the daily advice. The notification should be short; tapping it opens the full AI insight.

Example notification:

“AI Business Advice: Profit margin dropped 5% this week despite higher sales. Review discounts and cost changes.”

10. AI Data Pipeline

VisionFlow Database → KPI/Business Intelligence Layer → Alert & Trend Signals → AI Context Builder → AI Advisor → Insight Storage → Android App / Push Notification

The AI should receive structured, relevant business context rather than unrestricted direct database access. This improves consistency, security, cost control and auditability.

11. Recommended AI Context

- Current and historical sales KPIs.
- Profit and margin metrics.
- Expense trends.
- Receivables and aging.
- Inventory warnings and movement indicators.
- Branch/category/product performance where relevant.
- Existing business alerts from Phase 3.
- Relevant targets, budgets or thresholds when configured.

12. AI Cost & Performance Control

- Do not send the entire database to the AI.
- Pre-calculate KPIs and only send relevant context.
- Generate daily advice on a scheduled backend job where practical.
- Cache/store generated insights so every app open does not trigger a new AI request.
- Regenerate only when significant new data or conditions justify it.

13. Phase 4 Completion Criteria

1. AI can generate a daily business briefing from VisionFlow business intelligence data.
2. AI identifies meaningful trends, anomalies, risks and opportunities.
3. Each insight is supported by relevant business metrics.
4. Owner can open an insight and see the evidence behind it.
5. Needs Attention list prioritizes the most important issues.
6. Daily AI advice can be delivered through the Phase 3 push notification system.
7. AI history is available for previous recommendations.
8. AI does not directly modify any business data or execute operational actions.
9. Guardrails prevent unsupported or fabricated business claims.
10. AI processing is optimized through structured context, caching and controlled generation.

14. Final Phase Result

Result: AK VisionFlow Owner App becomes an intelligent executive companion. Instead of only showing numbers, it interprets the numbers, identifies what matters, explains why it may matter, and tells the owner what deserves attention.

Complete Owner App V1: Phase 1 Foundation → Phase 2 Executive Dashboard → Phase 3 Alerts & Notifications → Phase 4 AI Business Advisor.